Power BI Project - Dataset Evaluation through Power Query

Spotify + YouTube Music

Introduction to Spotify and YouTube Music

Spotify and YouTube Music are leading music streaming services used worldwide. Spotify, founded in Sweden, is known for its personalized playlists and podcast offerings. YouTube Music is owned by Google and leverages YouTube's vast library of songs, videos, and live performances. Both platforms operate on free and premium models, allowing users to stream music across devices.

📊 Project Overview

This project involves cleaning and preparing a Spotify and YouTube Music dataset using Power BI (Power Query) to ensure data accuracy and consistency. The dataset includes track details, artist information, engagement metrics, and audio features. Various data quality issues such as missing values, inconsistent formats, invalid entries, and duplicate records were identified and resolved. Power Query transformations were applied to standardize data, correct data types, and remove unnecessary columns. The cleaned dataset is now ready for reliable analysis and dashboard creation.

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1. Identify and Handle Missing Values:

Objective: Resolve missing and null values to preserve the accuracy and consistency of the dataset.

Questions:

- Examine the dataset for any missing values. Which columns contain null values?
- How should missing values in the Views and Likes columns be handled? Should they be filled with a default value, removed, or handled in another way? Justify your approach.

Identify columns with Missing data / Null values

This dataset contains missing values across multiple columns, including key, licensed, views, likes, comments, description, stream, official video, and several audio feature attributes such as valence and liveness.

Handling and Justification of Missing Values

- Two records contained null values in **key** column, and as all related data was missing, they were removed.
- The **Official Video** column has null values where videos are not officially published, and consequently, the related License information is also missing, as licensing applies only to official content.
  Since both fields are interdependent and essential for classification, records with null values were removed to ensure data consistency and analytical reliability.
- Invalid values in the **Views** column were first standardized to blanks. The data was then sorted by artist, and missing values were down filled using available view counts from the same artist to ensure consistent and logical imputation.
- For other numeric columns (**likes, comments, streams**), NULL values were replaced with artist-level averages using group by .
  Because songs by the same artist usually show similar engagement levels, artist-level averaging provides realistic values without distorting the data.

2. Fix Irregularities in Merged Columns:

Objective: Separate merged columns into their original components for accuracy.

Questions:

- The Spotify Info and Youtube Info columns contain merged data separated by delimiters. Split these columns back into their original components. What are the original components, and how can you ensure that the split data is clean and accurate?
- After splitting, remove any unnecessary delimiters or prefixes/suffixes that do not belong.

Components of the Spotify_Info and YouTube_Info and methods to seperate them

The **Spotify Info** column contains both the **Spotify album link** and the **Spotify track ID.** These can be separated by splitting the column using the "|" delimiter, resulting in two new columns: Spotify Album Link and Spotify Track Id.	The **YouTube Info** column contains the **YouTube track link** and **track title**.The column can be split accordingly to create youtube track link and youtube track title.

Ensuring Data Cleanliness and Accuracy After Column Splitting

Verify the functionality of Spotify links, confirm the accuracy of track IDs, and ensure YouTube video titles are correctly and consistently extracted.

3. Correct Case Sensitivity and Naming Conventions:

Objective: Ensure all column names and data entries follow consistent casing and standardized naming conventions.

Questions:

- The column names have inconsistent case sensitivity (some are uppercase, others lowercase). Standardize all column names to follow a consistent format (e.g., all lowercase with underscores).
- Fix any data entries where case sensitivity might affect consistency (e.g., artist names or track titles). Ensure that the Artist and Track columns are formatted consistently.

Standardizing Column Name Case and Formatting

Column names should be capitalized in title case (e.g., "Spotify Info" instead of "spotify_info"
Text entries such as artist names and track titles should be capitalized consistently.

Ensuring Consistent Formatting

To ensure all column names follow a consistent format, we transformed the case by capitalizing each word for both the column headers and the content within the columns. This approach improves readability, maintains uniformity, and ensures consistent formatting across the dataset

4. Remove or Handle Irrelevant Columns

Objective: Eliminate or appropriately manage columns that do not contribute to the analysis.

Questions:

- Identify and remove any irrelevant or randomly generated columns that do not provide useful information for analysis. Which columns should be removed, and why?
- If any random data exists in relevant columns, clean or remove those entries.

Identification and Removal of Irrelevant Columns

The columns '**Random Column 1**','**Random Column 2**', **unnamed: 0** , were identified as randomly generated and irrelevant to the dataset.

Since they do not contribute any meaningful information for analysis, these columns were removed to maintain dataset clarity and usability

5. Handle Inconsistent Data Types:

Objective: Ensure all columns have appropriate and consistent data types for accurate analysis.

Questions:

- Some columns that should be numeric (e.g., Danceability, Energy) are stored as text. Convert these columns back to numeric format. What steps would you take to identify and fix any issues that arise during this conversion?
- Ensure that all numeric columns are in the correct format and handle any non-numeric values or anomalies.

Data Type Validation and Numeric Conversion

The **Views, Danceability, and Energy** columns were initially stored as text and were corrected by converting them into appropriate numeric (decimal) data types. Before conversion, non-numeric values, blanks, and invalid entries were identified and handled to prevent errors.
The columns were then transformed into numeric formats, and validation checks were performed to ensure consistency and accuracy for analysis and calculations.

6. Address and Fix Invalid Data Entries:

Objective: Identify and correct erroneous or invalid data to maintain dataset accuracy and reliability.

Questions:

- Check the Views column for any entries labeled as "invalid_data" or any other incorrect values. Replace these entries and justify your method.
- Ensure that all values in the Album column are correctly labeled and that there are no numeric entries or irrelevant data.

Views Data Validation and Correction

The **Views** column was checked for values like “invalid_data”, which were replaced with NULL for consistency. These missing values were then filled using **artist-level averaging**, as songs by the same artist typically have similar view counts. This method ensures realistic imputation without introducing bias.

Album Column Validation and Consistency Check

The **Album** column shows 100% valid values with no errors or empty entries, all records accurately represent album titles.
The presence of numbers or special characters is valid in this context, as many album names legitimately include them. Modifying or removing such values could lead to loss of meaningful information, therefore the column was left unchanged to preserve data integrity and originality.

7. Check for and Remove Duplicate rows

Objective: Identify and eliminate duplicate records to ensure data uniqueness and integrity.

Question:

- Identify and remove any duplicate rows in the dataset. How can you ensure that the remaining data is unique and accurate?

Duplicate detection was conducted using multiple key columns to identify unique records. While some values repeated, they reflected distinct entries with meaningful differences across artists, channels, or platforms.
As no exact duplicates were found, all records were retained, ensuring the dataset remains complete and reliable for analysis.

8. Reorder and Rename Columns for Clarity:

Questions:

- Reorder the columns in a logical sequence to improve the dataset's readability and usability. What order makes the most sense for this dataset?
- Rename columns where necessary to ensure that their names clearly reflect the data they contain.

Reordering the columns as

Starting from identifying information, then categorical/qualitative data, followed by quantitative/audio features, and finally engagement metrics.

Track → Artist → Album → Album Type → Spotify Album Link → Spotify Track Id → YouTube track link → Youtube track title → Official Video → Channel → Description → Licensed → Duration Ms → Key → Tempo → Danceability → Energy → Valence → Liveness → Speechiness → Acousticness → Instrumentalness → Loudness → Views → Likes → Comments → Stream

Identify first: Track, Artist, Album.

- Links and official info: Helps track sources.
- Audio features next: Grouping all musical/audio attributes together makes analysis easier.
- Engagement last: Views, likes, comments, and streams naturally follow after the track info.

Data Cleaning and Validation Summary

After performing data cleaning and validation, the dataset is now complete, consistent, and ready for analysis.
Missing or invalid values were appropriately handled using methods such as artist-level averaging or removal of irrecoverable entries.
Column names and data entries were standardized for consistency, and numeric fields were corrected to proper formats.
Overall, the dataset maintains integrity and reliability, ensuring accurate and meaningful analytical results..

Presented By	Divya Sharma
LinkedIn	www.linkedin.com/in/divya9916
GitHub	github.com/Divya9916